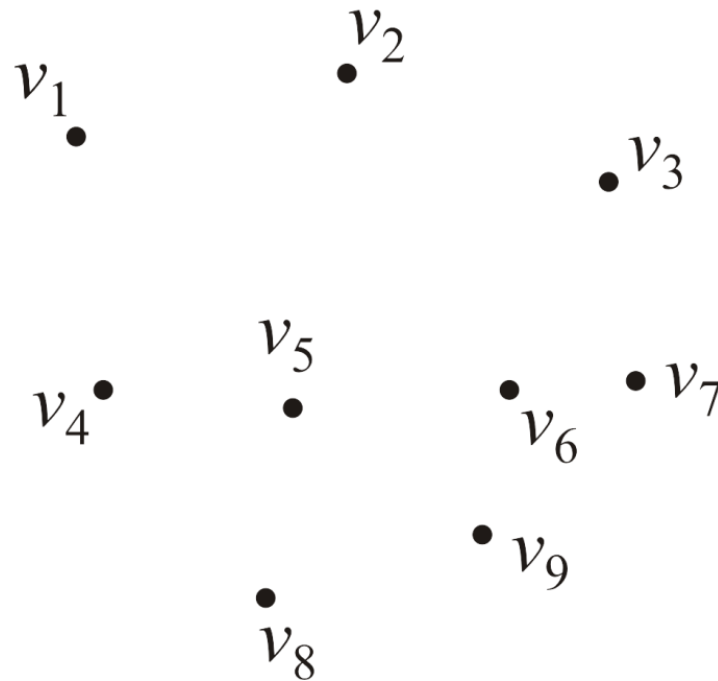


Data Structures

23. Graphs

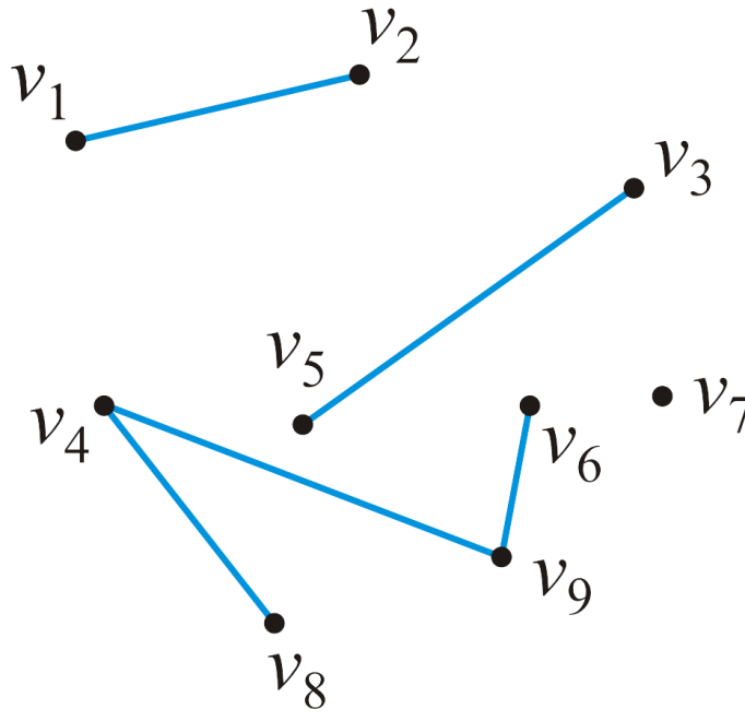
Graphs

- Consider this collection of vertices
 - $V = \{v_1, v_2, \dots, v_9\}$
 - Where $|V| = n$



Graphs

- Associated with these vertices are $|E| = 5$ edges
 - $E = \{\{v_1, v_2\}, \{v_3, v_5\}, \{v_4, v_8\}, \{v_4, v_9\}, \{v_6, v_9\}\}$
- Pair $\{v_j, v_k\}$ indicates following relations
 - Vertex v_j is adjacent to vertex v_k
 - Vertex v_k is adjacent to vertex v_j

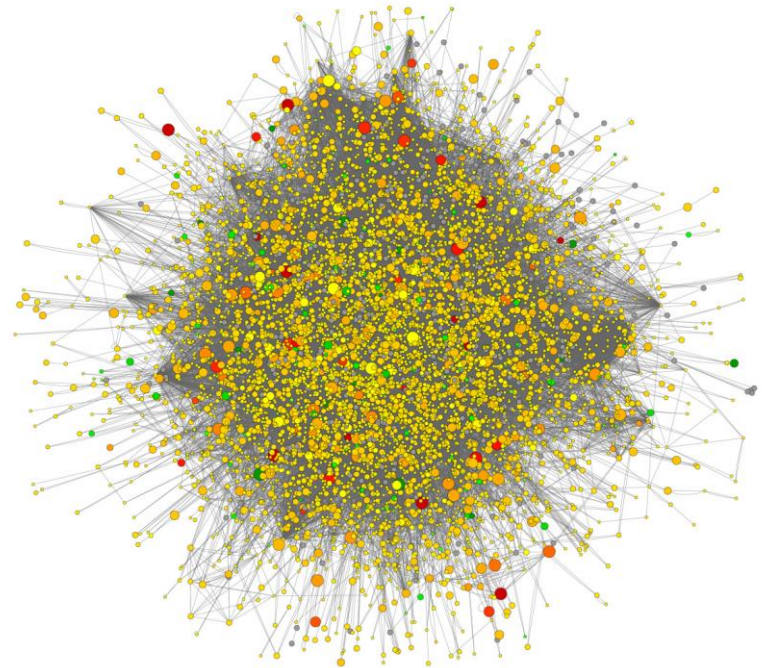


Applications Of Graphs

- Driving Map
 - Vertex = Intersection, destinations
 - Edge = Road
- Airline Traffic
 - Vertex = Cities serviced by the airline
 - Edge = Flight exists between two cities
- Computer networks
 - Vertex = Server nodes, end devices, routers
 - Edge = Data link

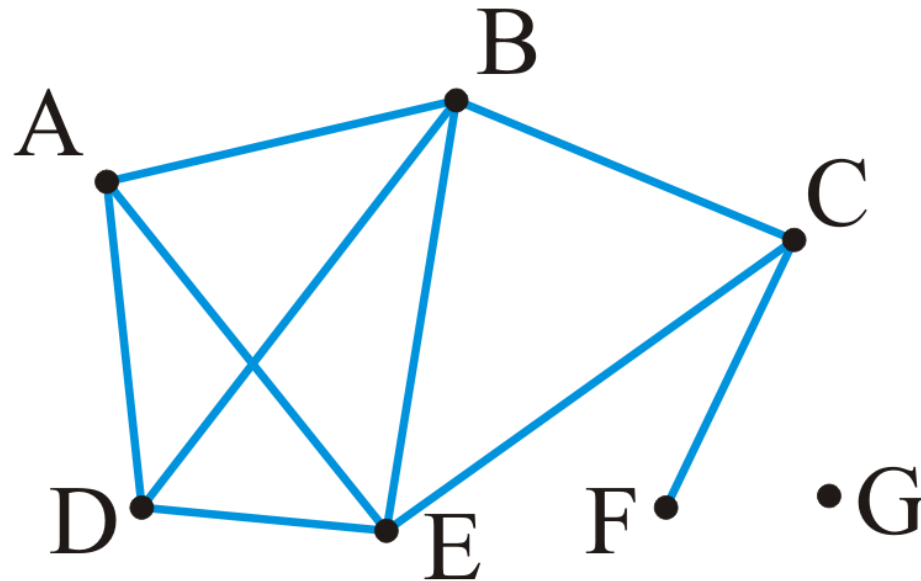
Applications Of Graphs

- Many real-world applications concern large graphs
- Web document graph - 1 trillion webpages
 - Vertex = Webpage
 - Edge = Hyperlink
- Social networks - 1.3 billion users
 - Vertex = Users
 - Edge = Friendship relation



Undirected Graphs – Example

- Given $|V| = 7$ vertices and $|E| = 9$ edges
 - $V = \{A, B, C, D, E, F, G\}$
 - $E = \{\{A, B\}, \{A, D\}, \{A, E\}, \{B, C\}, \{B, D\}, \{B, E\}, \{C, E\}, \{C, F\}, \{D, E\}\}$

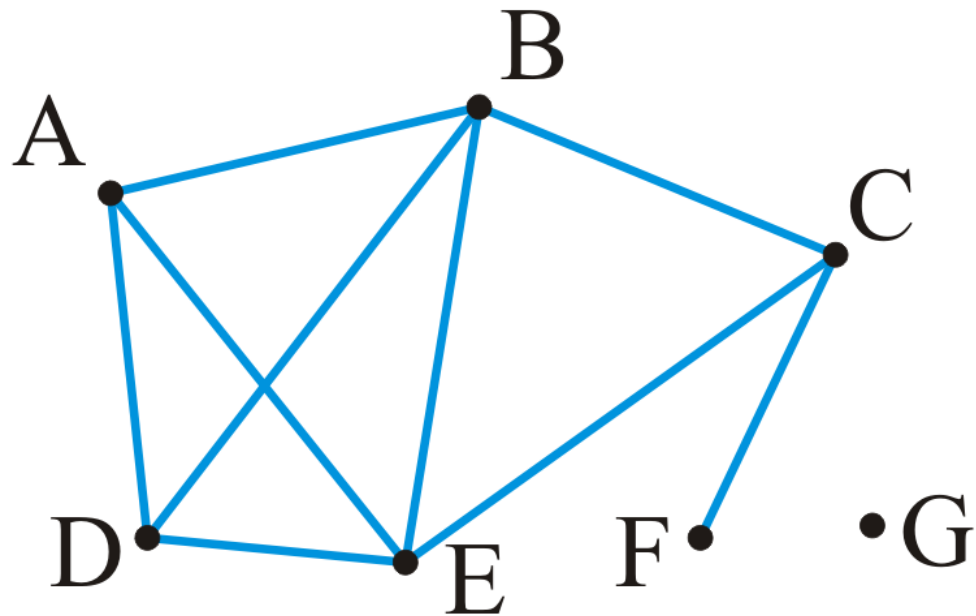


Undirected Graphs – Definition

- An **undirected Graph** is defined as $G=(V, E)$ consisting of
 - Set V of **vertices**: $V = \{v_1, v_2, \dots, v_n\}$
 - Number of vertices is denoted by $|V| = n$
 - Set E of unordered pairs $\{v_i, v_j\}$ termed **edges**
 - Edges connect the vertices
- **Maximum number of edges** in an undirected graph is $O(|V|^2)$
$$|E| \leq \binom{|V|}{2} = \frac{|V|(|V|-1)}{2} = O(|V|^2)$$
 - Assumption: A vertex is never adjacent to itself
 - For example, $\{v_1, v_1\}$ will not define an edge
- Many data structures can implement abstract undirected graphs
 - Adjacency matrices, Adjacency lists

Degree

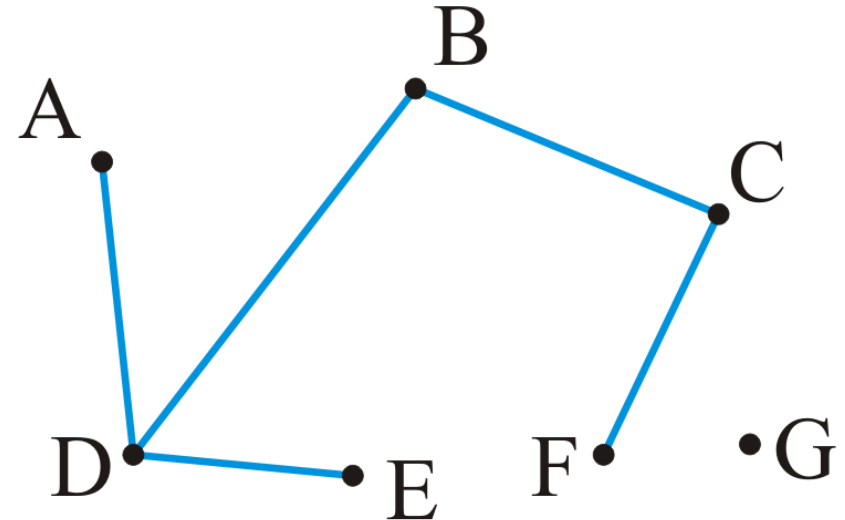
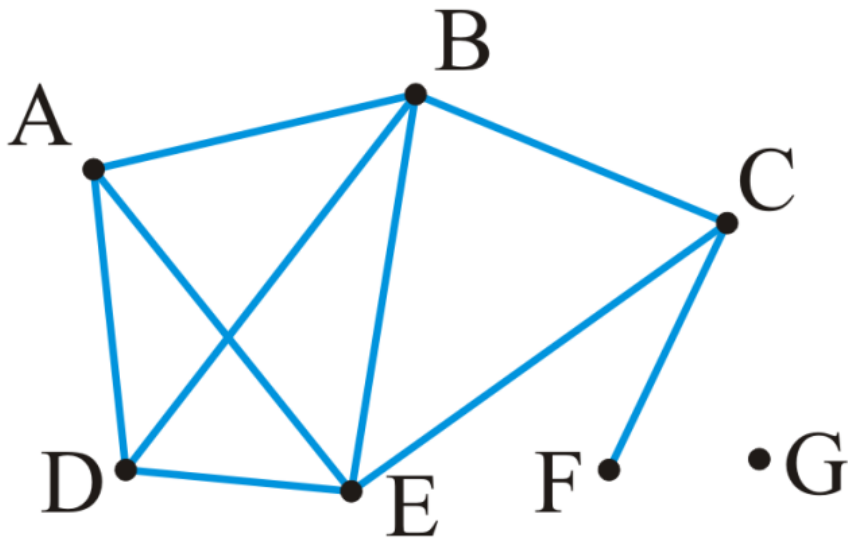
- Degree of a vertex is defined as the number of adjacent vertices
 - $\text{degree}(A) = \text{degree}(D) = \text{degree}(C) = 3$
 - $\text{degree}(B) = \text{degree}(E) = 4$
 - $\text{degree}(F) = 1$
 - $\text{degree}(G) = 0$



- Vertices adjacent to a given vertex are its neighbors

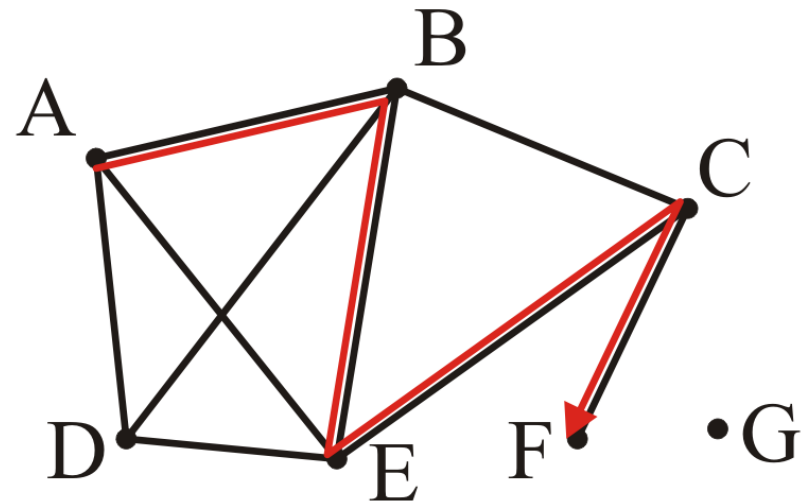
Subgraph

- A sub-graph of a graph G is defined by
 - Subset of the vertices
 - Subset of the edges that connected the subset of vertices in the original graph
- Every graph is a subgraph of itself



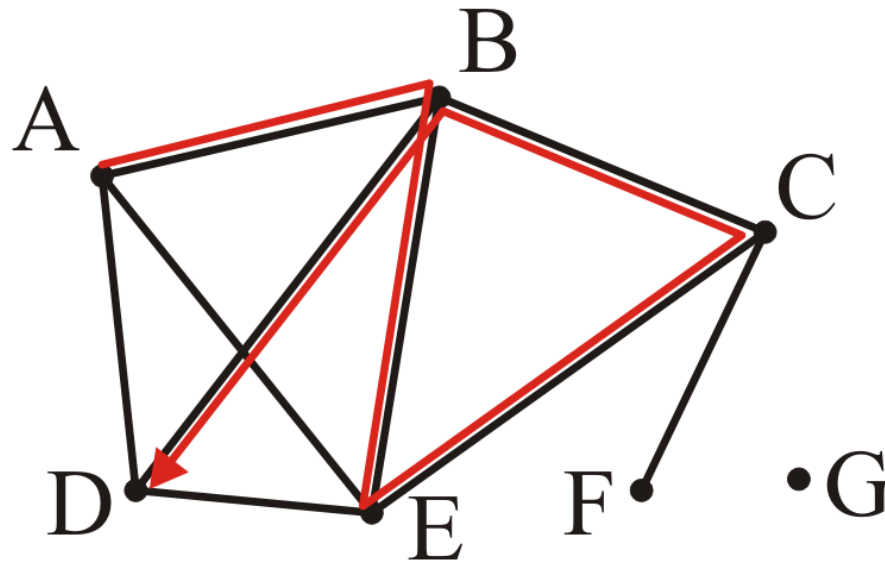
Path

- Path in an undirected graph is an ordered sequence of vertices
 - Consecutive vertices are connected through edges
- Path from vertex θ to vertex k is $(v_\theta, v_1, v_2, \dots, v_k)$
 - where $\{v_{j-1}, v_j\}$ is an edge for $j = 1, \dots, k$
- Length of a path is equal to the number of edges
- Example: Path from A to F
 - Path: (A, B, E, C, F)
 - Length of the path is 4



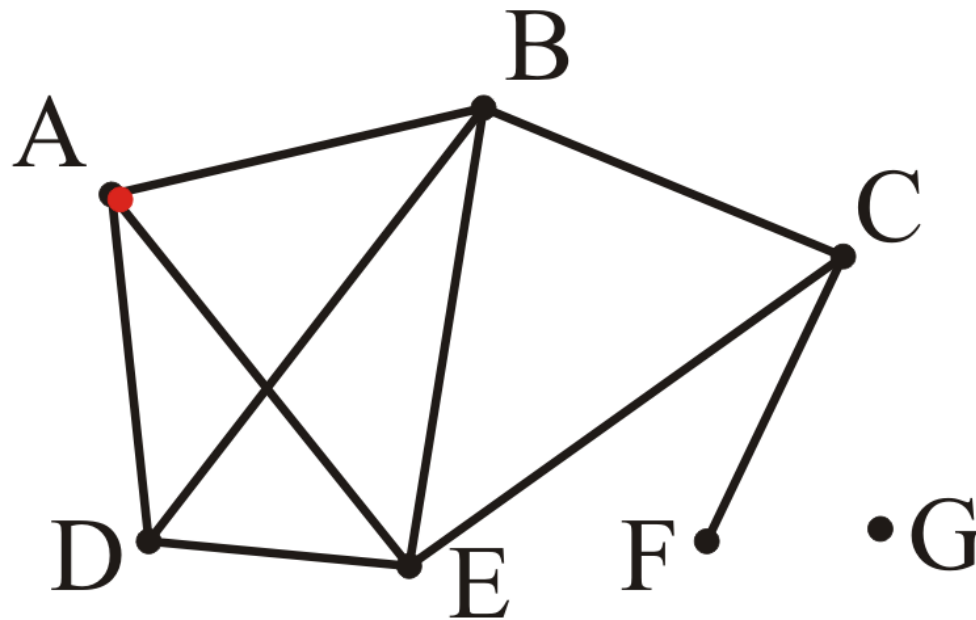
Path – Example

- Path of length 5: (A, B, E, C, B, D)
 - Repetition of vertex B



Path – Example

- A trivial path of length 0: (A)

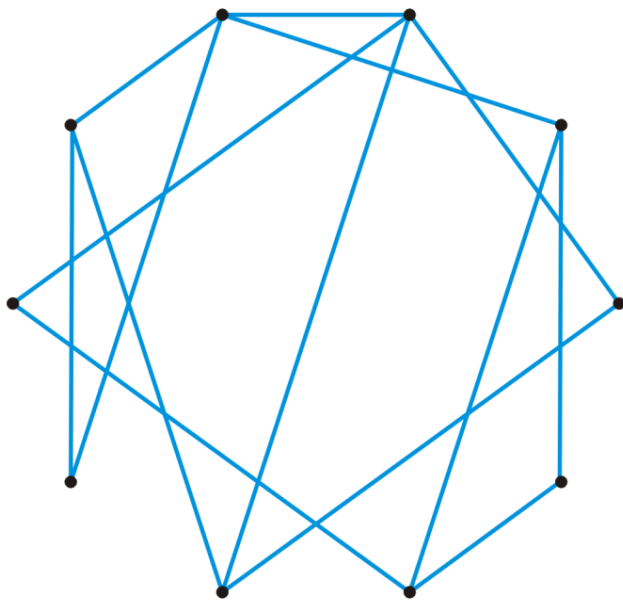


Simple Path

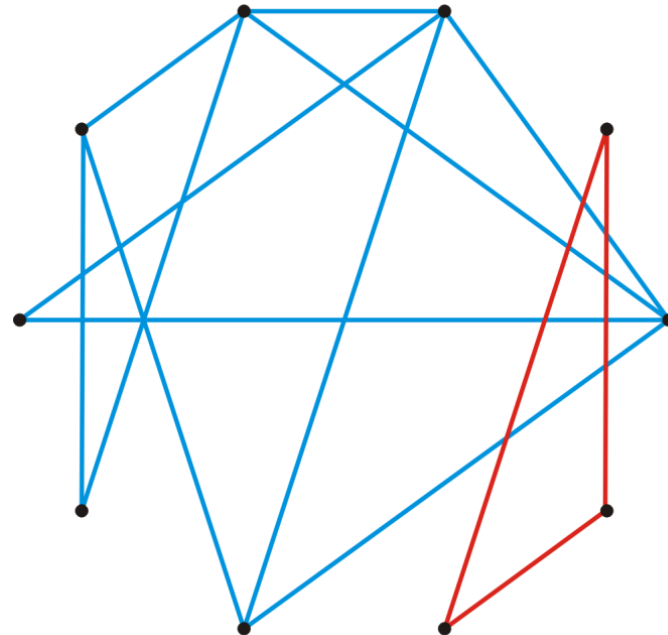
- A simple path has no repetitions other than perhaps the first and last vertices
- A simple cycle is a simple path of at least two vertices with the first and last vertices equal
 - Note: these definitions are not universal
- A loop is an edge from a vertex onto itself

Connectedness

- Two vertices v_i, v_j are said to be connected if there exists a path from v_i to v_j
- A graph is connected if there exists a path from every vertex to every other vertex



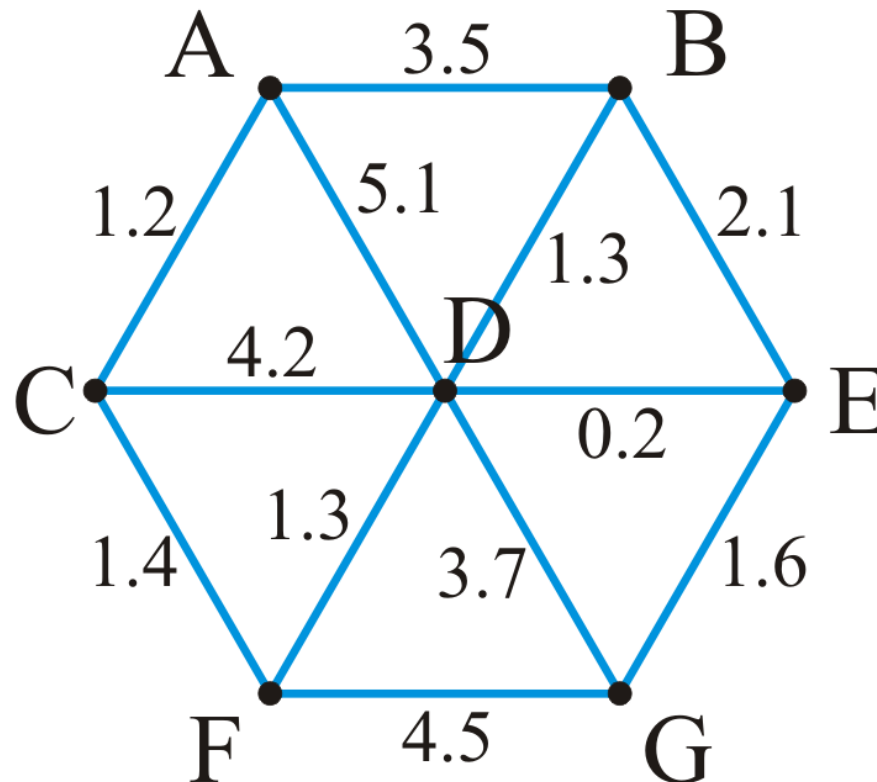
A connected graph



An unconnected graph

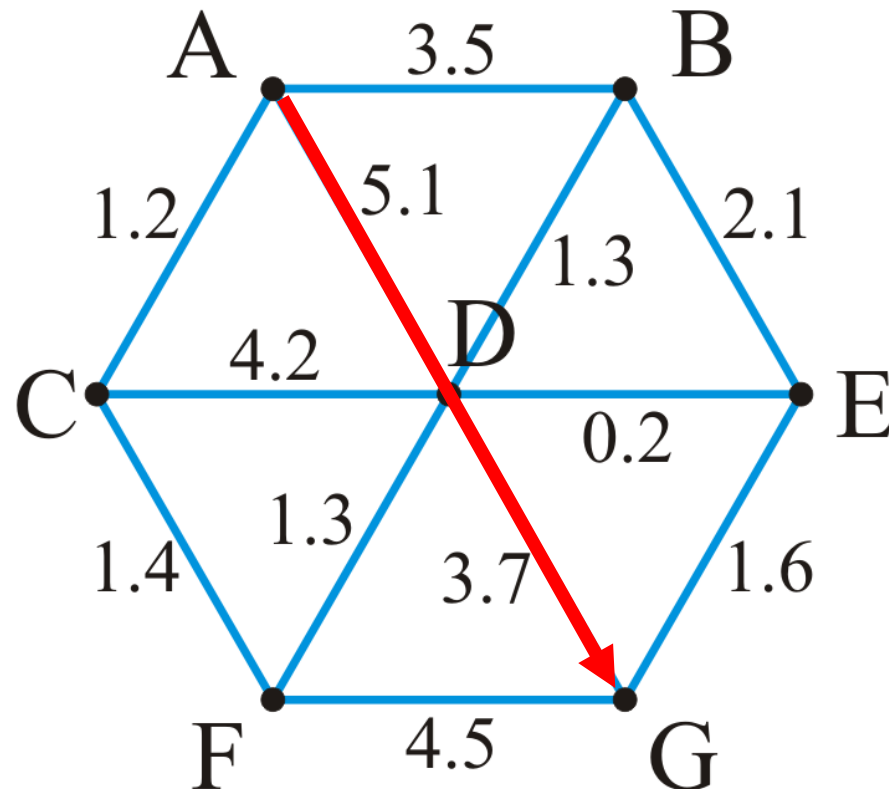
Weighted Graphs

- A weight may be associated with each edge in a graph
 - This could represent distance, energy consumption, cost, etc.
 - Such a graph is called a weighted graph
- Pictorially, we will represent weights by numbers next to the edges



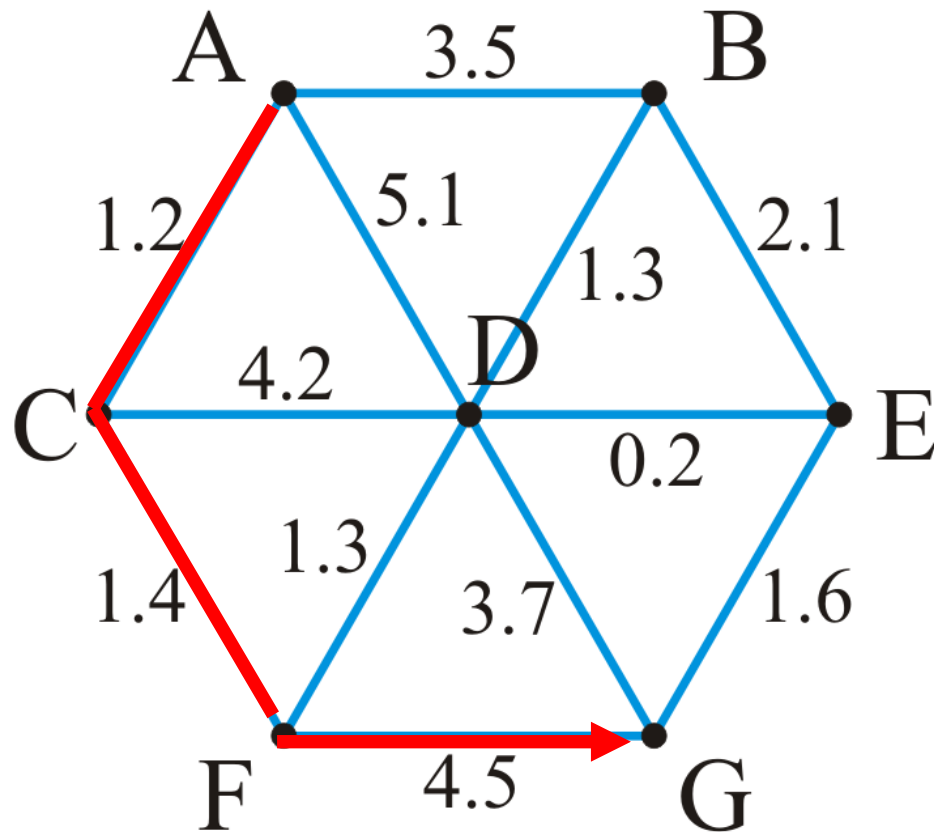
Weighted Graphs

- Length of a path within a weighted graph is the sum of all of the edges which make up the path
- The length of the path (A, D, G) in the following graph is 8.8



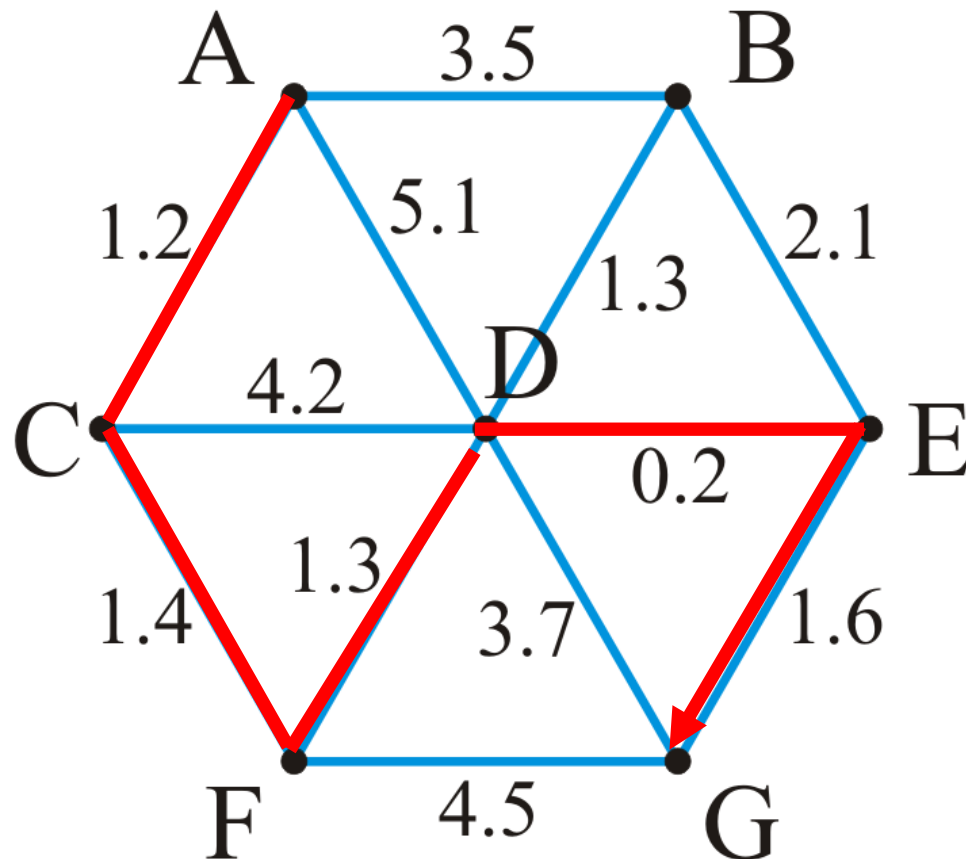
Weighted Graphs – Example

- Different paths may have different weights
 - Another path is (A, C, F, G) with length $1.2 + 1.4 + 4.5 = 7.1$



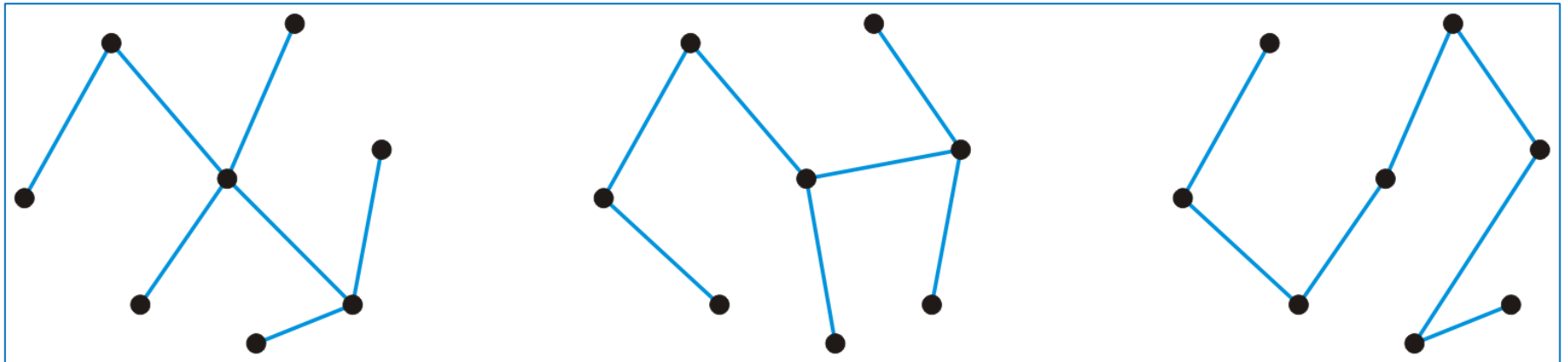
Weighted Graphs – Example

- Find the shortest path between two vertices A and G
- Shortest path is (A, C, F, D, E, G) with length 5.7



Trees

- A graph is a tree if it satisfies the following two conditions
 - Graph is connected
 - There is a unique path between any two vertices



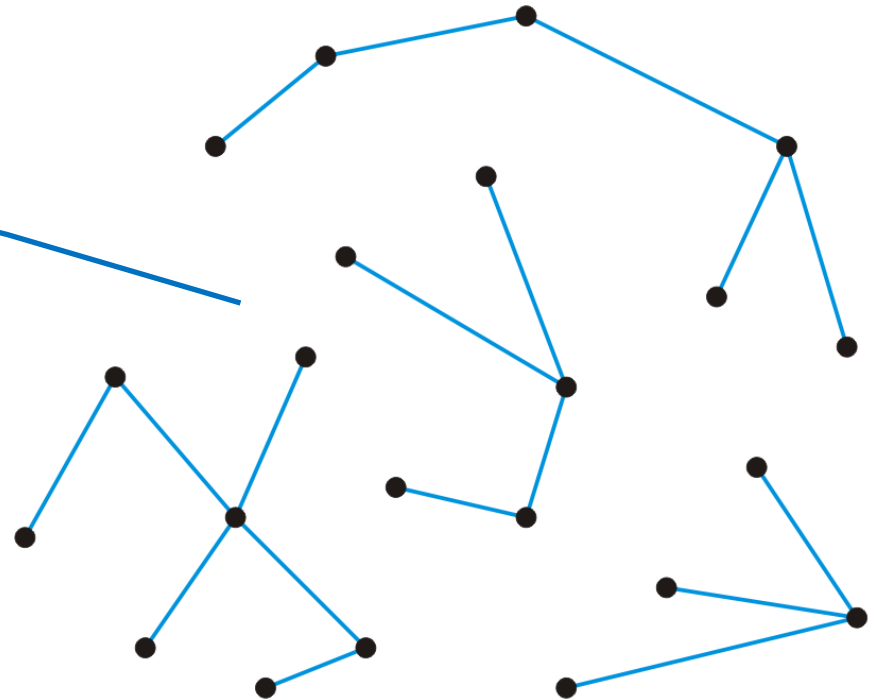
Three trees on
same 8 vertices

- Consequences
 - The number of edges is $|E| = |V| - 1$
 - The graph is acyclic, that is, it does not contain any cycles
 - Adding one more edge must create a cycle
 - Removing any one edge creates two disjoint non-empty sub-graphs

Forest

- A forest is any graph that has no cycles
- Consequences
 - The number of edges is $|E| < |V|$
 - The number of trees is $|V| - |E|$
 - Removing any one edge adds one more tree to the forest

- Forest with 22 vertices and 18 edges
- Four trees

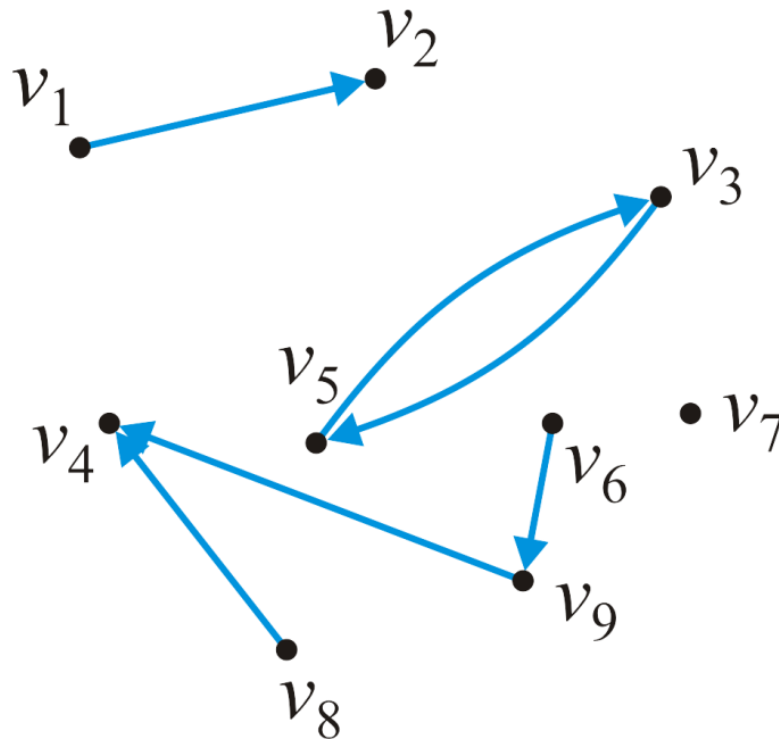


Directed Graphs

- In a directed graph, the edges on a graph are associated with a direction
 - Edges are ordered pairs (v_j, v_k) denoting a connection from v_j to v_k
 - The edge (v_j, v_k) is different from the edge (v_k, v_j)
- Streets are directed graphs
 - In most cases, you can go two ways unless it is a one-way street

Directed Graphs

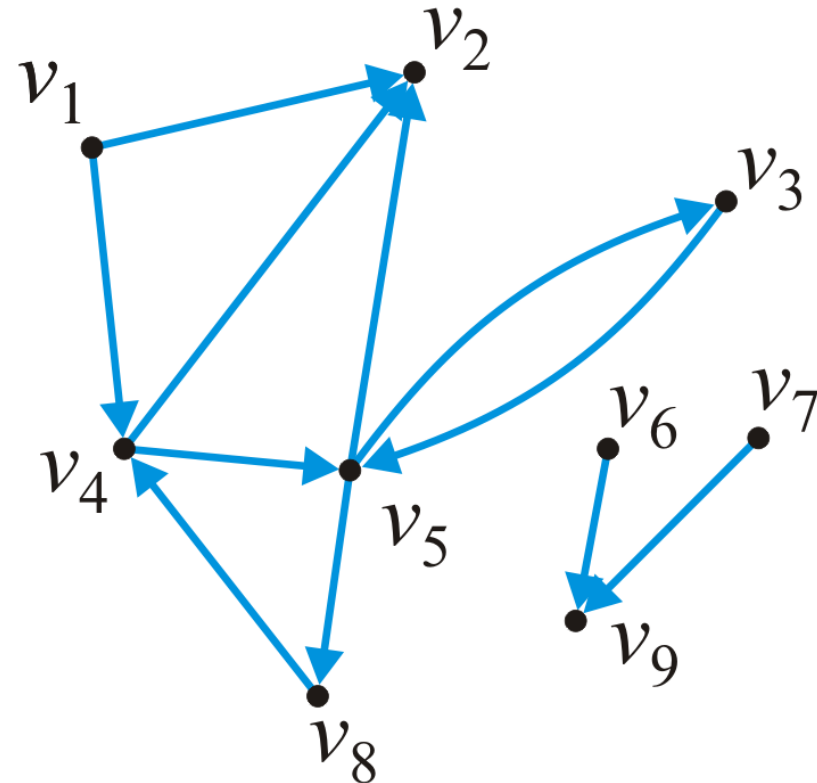
- Given our graph of nine vertices $V = \{v_1, v_2, \dots, v_9\}$
 - These six pairs (v_j, v_k) are directed edges
 - $E = \{(v_1, v_2), (v_3, v_5), (v_5, v_3), (v_6, v_9), (v_8, v_4), (v_9, v_4)\}$



In and Out Degree

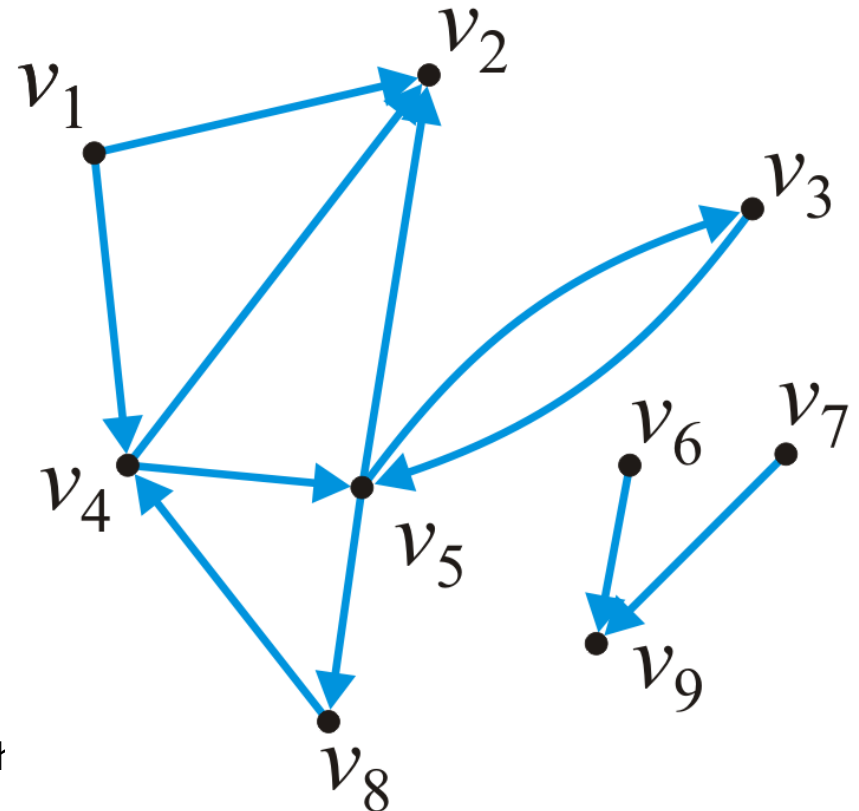
- Degree of a vertex must be modified to consider both cases:
 - **Out-degree** of a vertex is the number of vertices which are adjacent to the given vertex
 - Number of outgoing edges
 - **In-degree** of a vertex is the number of vertices which this vertex is adjacent to
 - Number of incoming edges

- In this graph:
 - $\text{In-degree}(v_1) = 0$ $\text{out-degree}(v_1) = 2$
 - $\text{In-degree}(v_5) = 2$ $\text{out-degree}(v_5) = 3$



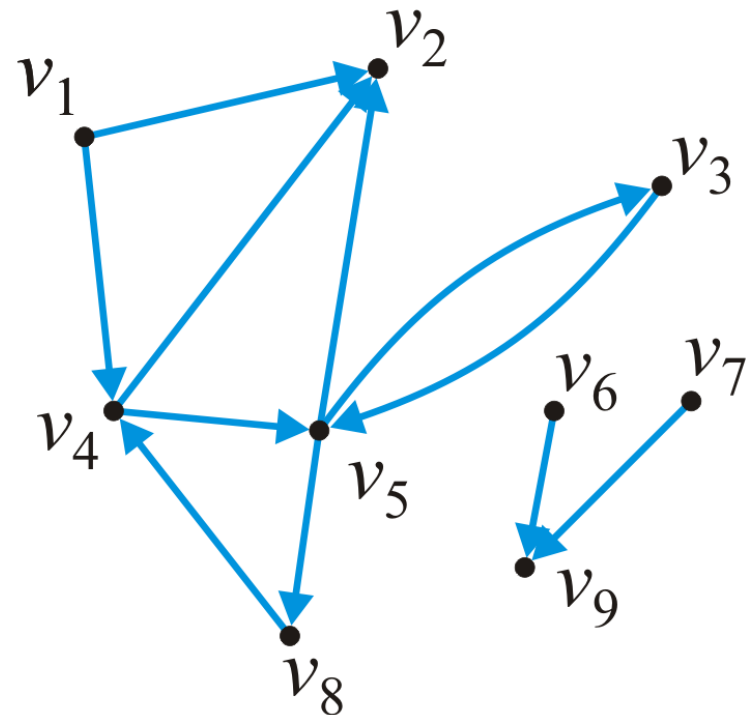
Path

- A path in a directed graph is an ordered sequence of vertices
 - $(v_0, v_1, v_2, \dots, v_k)$
 - where (v_{j-1}, v_j) is an edge for $j = 1, \dots, k$
- A path of length 5 in this graph is
 - $(v_1, v_4, v_5, v_3, v_5, v_2)$
- A simple cycle of length 3 is
 - (v_8, v_4, v_5, v_8)



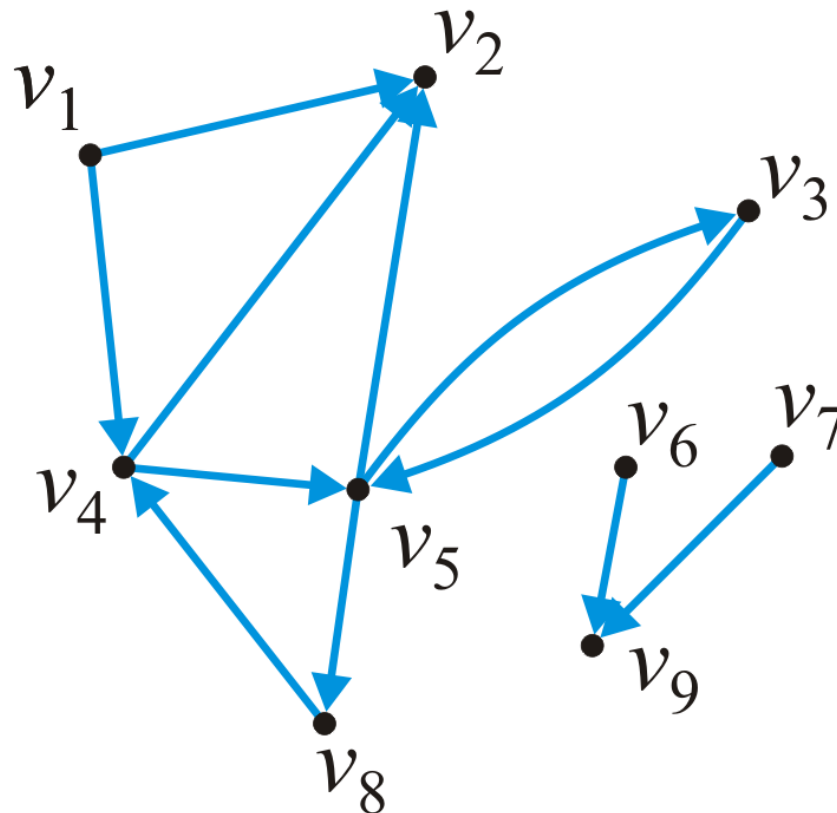
Connectedness

- Two vertices v_j , v_k are said to be connected if there exists a path from v_j to v_k
 - A graph is **strongly connected** if there exists a directed path between any two vertices
 - A graph is **weakly connected** if there exists a path between any two vertices that ignores the direction



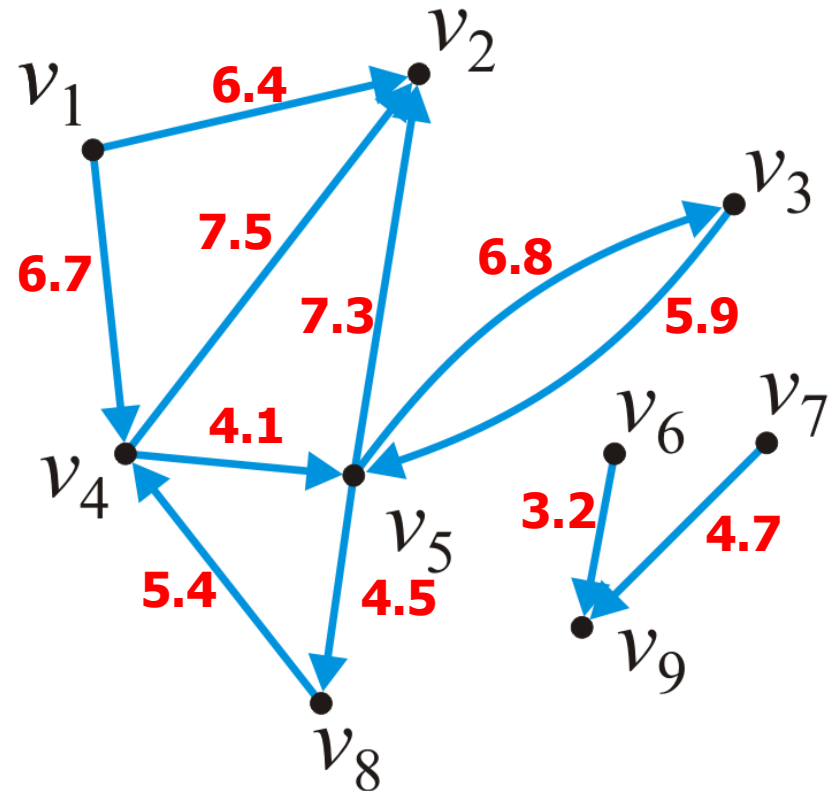
Connectedness – Example

- The sub-graph $\{v_3, v_4, v_5, v_8\}$ is strongly connected
- The sub-graph $\{v_1, v_2, v_3, v_4, v_5, v_8\}$ is weakly connected



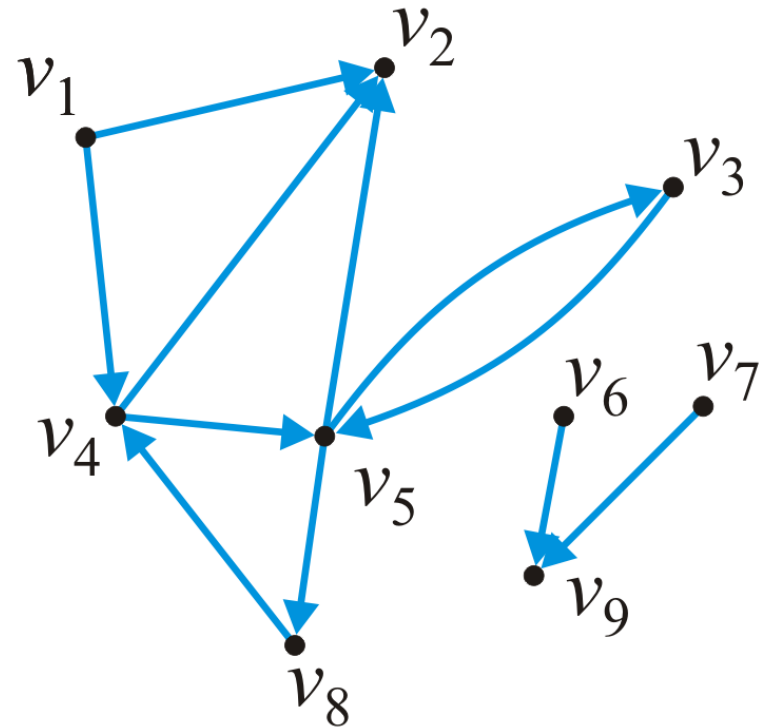
Weighted Directed Graphs

- Each edge is associated with a value
- If both (v_j, v_k) and (v_k, v_j) are edges
 - It is not required that they have the same weight



Representation

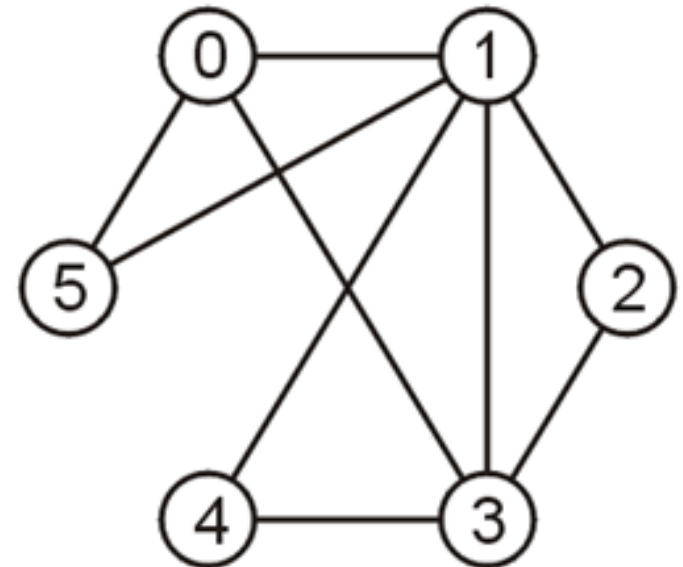
- How do we store the adjacency relations?
 - Adjacency matrix
 - Adjacency list



Adjacency Matrix

- Two dimensional matrix of size $n \times n$ where $n = |V|$
- $a[i, j] = 0$ (F) if there is no edge between vertices v_i and v_j
- $a[i, j] = 1$ (T) if there is an edge between vertices v_i and v_j
- Adjacency matrix of undirected graphs is symmetric
 - $a[i, j] = a[j, i]$

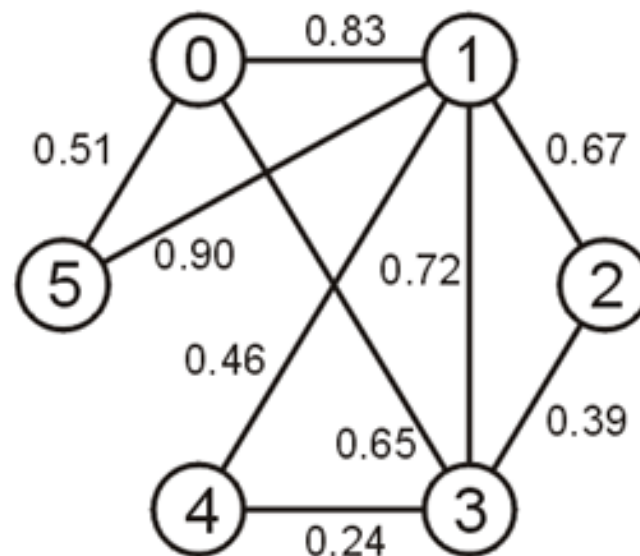
	0	1	2	3	4	5
0	F	T	F	T	F	T
1	T	F	T	T	T	T
2	F	T	F	T	F	F
3	T	T	T	F	T	F
4	F	T	F	T	F	F
5	T	T	F	F	F	F



Adjacency Matrix – Weighted Graph

- The matrix entry $[j, k]$ is set to the weight of the edge (v_j, v_k)
- How to indicate absence of an edge in the graph?

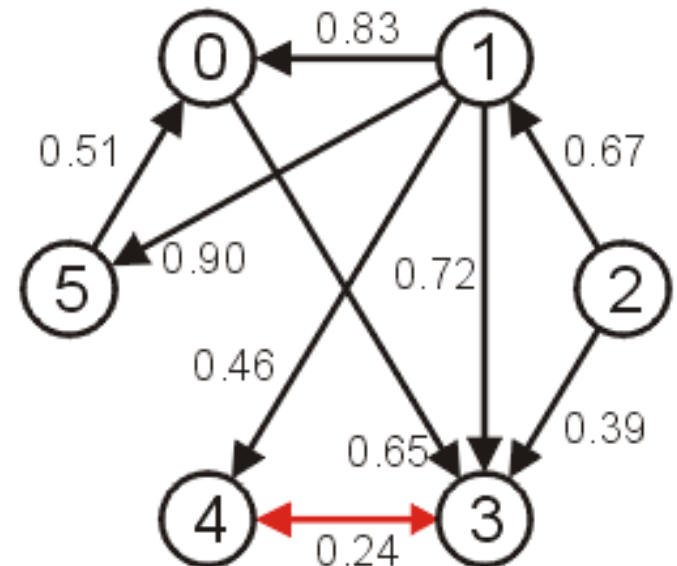
	0	1	2	3	4	5
0		0.83		0.65		0.51
1	0.83		0.67	0.72	0.46	0.90
2		0.67		0.39		
3	0.65	0.72	0.39		0.24	
4		0.46		0.24		
5	0.51	0.90				



Adjacency Matrix – Directed Graph

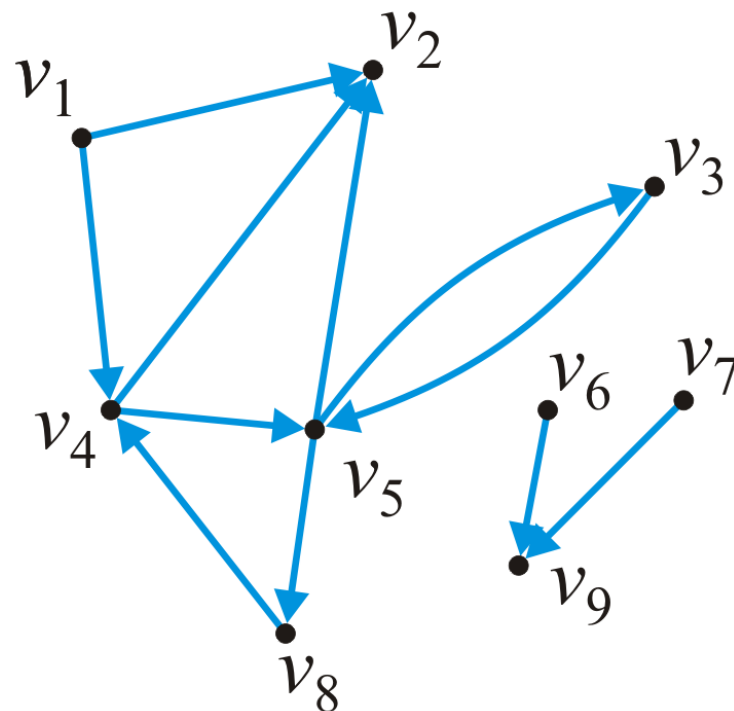
- For directed graph the matrix would not necessarily be symmetric

	0	1	2	3	4	5
0				0.65		
1	0.83			0.72	0.46	0.90
2		0.67		0.39		
3					0.24	
4				0.24		
5	0.51					



Adjacency Matrix – Analysis

	1	2	3	4	5	6	7	8	9
1		1		1					
2									
3					1				
4		1			1				
5		1	1					1	
6									1
7									1
8				1					
9									

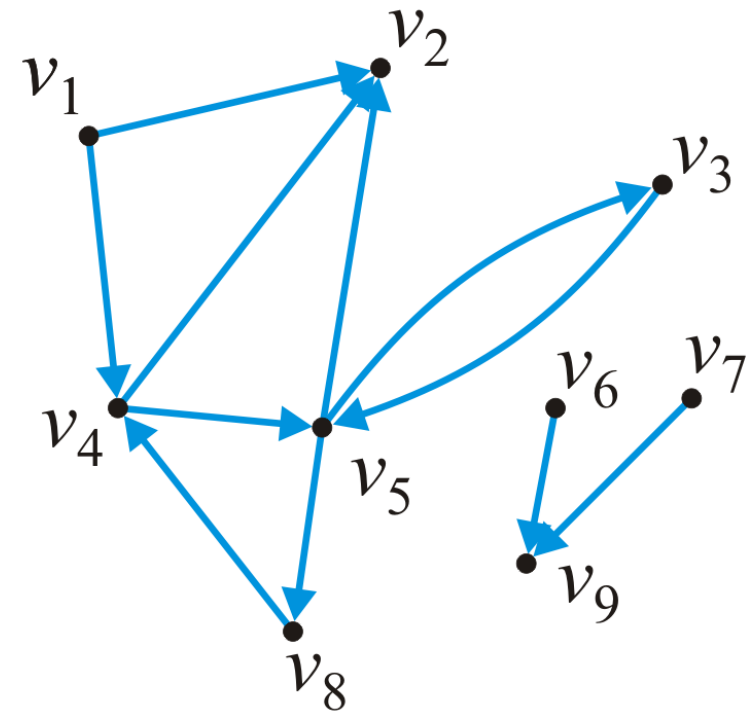


- Requires memory : $O(|V|^2)$
- Determining if v_j is adjacent to v_k : $O(1)$
- Finding all neighbors of v_j : $O(|V|)$

Adjacency Matrix – Problem

- Very sparsely populated
 - Out of 81 cells only 11 are 1 (or T)

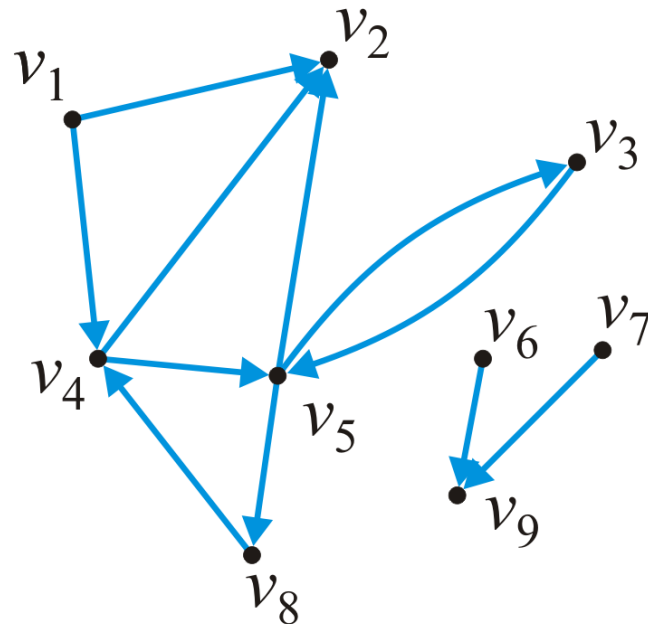
	1	2	3	4	5	6	7	8	9
1		1		1					
2									
3					1				
4		1			1				
5		1	1					1	
6									1
7									1
8				1					
9									



Adjacency List

- Each vertex is associated with a list of its neighbors
 - A vertex w is inserted in the list for vertex v if edge (v, w) exists

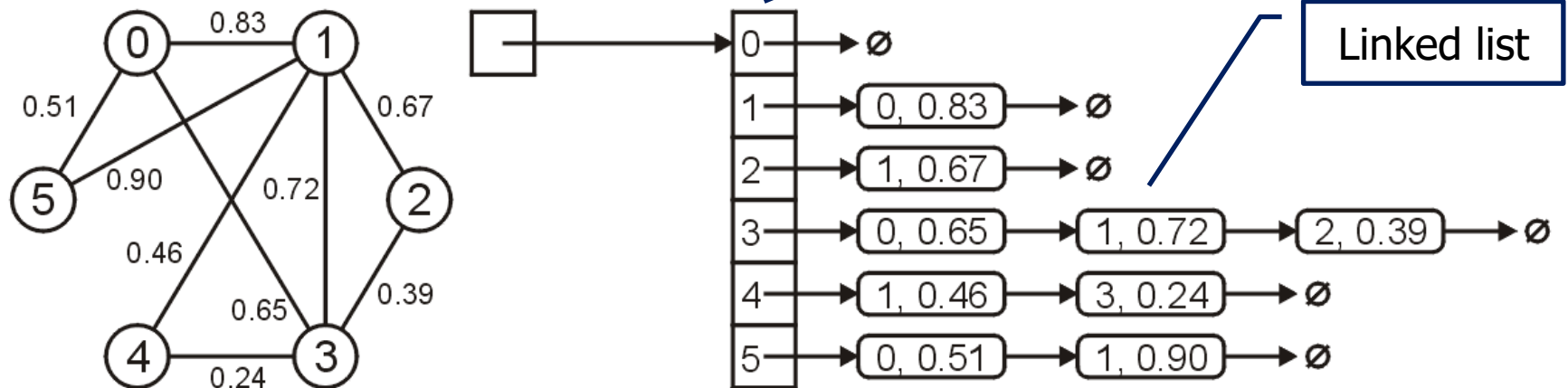
1	• → 2 → 4
2	•
3	• → 5
4	• → 2 → 5
5	• → 2 → 3 → 8
6	• → 9
7	• → 9
8	• → 4
9	•



- Requires memory : $O(|V| + |E|)$

Adjacency List – Weighted Graphs

- An adjacency list for a weighted graph contains two elements
 - First element for the vertex
 - Second element for the weight of that edge



- When the vertices are identified by a name (i.e., string)
 - Hash-table of lists is used to implement the adjacency list

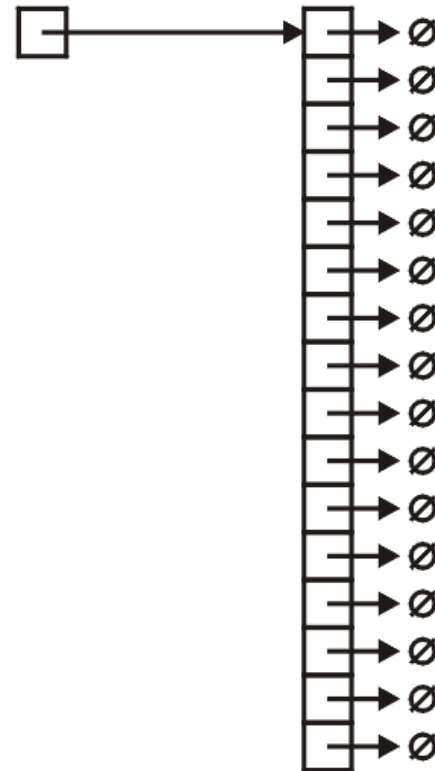
Adjacency List – Implementation

- Node to store adjacent vertex and weight of the edge

```
class SingleNode {  
    private:  
        int adjacent_vertex;  
        double edge_weight;  
        SingleNode * next_node;  
};
```

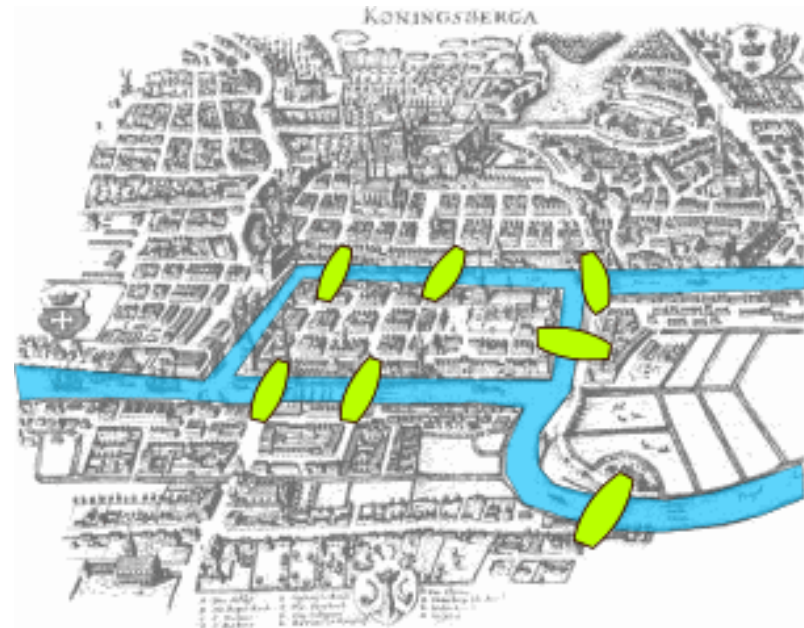
- Define and create Array

```
SingleNode * array;  
array = new SingleNode[16];
```



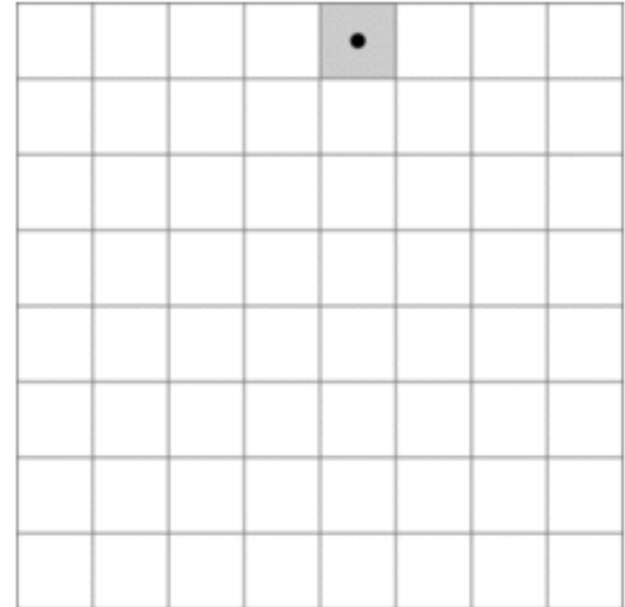
Graph Problems – Euler Tour

- A sequence of vertices that traverse all edges in the graph exactly once
 - Leonhard Euler in 1736
 - Laid the foundations of graph theory
- **Problem:** To devise a walk through the city that would cross each of the seven bridges of Königsberg once and only once
 - Euler proved problem has no solution



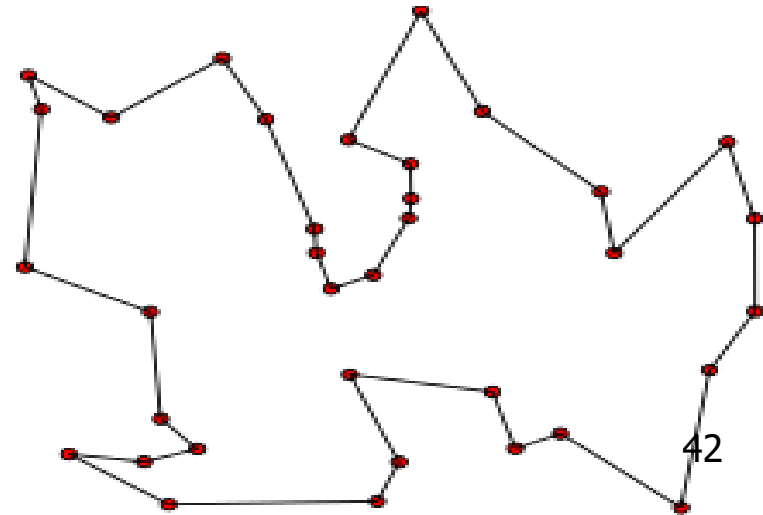
Graph Problems – Hamiltonian Cycle

- Is there a simple cycle that connects all vertices in the graph
 - NP-Complete
- Knight's Tour of chessboard
 - A sequence of knight's moves
 - Visit every square of a chessboard precisely once
 - Returns to its initial square



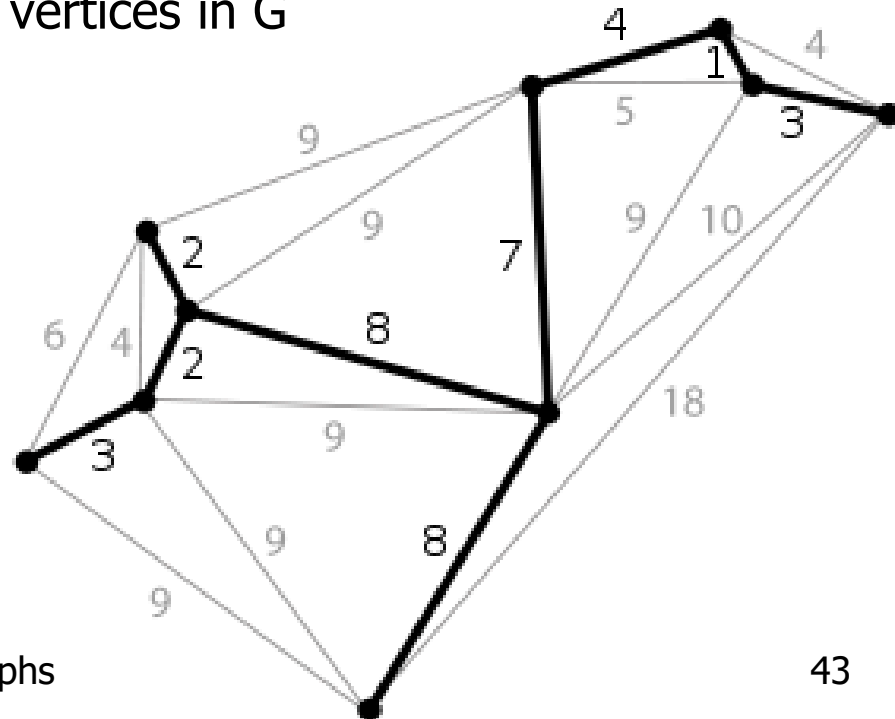
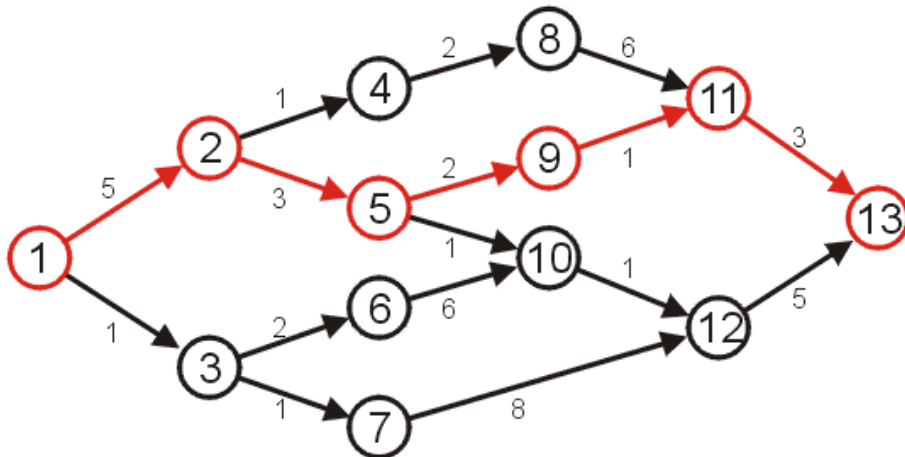
Graph Problems – Traveling Salesman

- A salesman wishes to
 - Visit a number of towns, and then
 - Return to his starting town
- Given the travelling times between towns, how should the travel be planned, so that:
 - He visits each town exactly once, and
 - He travels in as short time as possible
- **Problem:** Given a weighted graph G , provide shortest cycle that contains all vertices in G
 - NP-Hard problem



Graph Problems – Others

- **Minimum-cost spanning tree**
 - Given a weighted graph G , determine a spanning tree with minimum total edge cost
- **Single-source shortest path**
 - Given a weighted graph G and a source vertex v in G , determine the shortest paths from v to all other vertices in G



Any Question So Far?



Undirected Graphs – Definition

- An **undirected Graph ADT** is a **collection of vertices**
 - $V = \{v_1, v_2, \dots, v_n\}$
- The number of vertices is denoted by $|V| = n$
- Associated with this is a **collection E of unordered** pairs $\{v_i, v_j\}$ termed **edges** which connect the vertices
- **Maximum number of edges** in an undirected graph is $O(|V|^2)$
 - Assumption: A vertex is never adjacent to itself (for this course)
 - For example, $\{v_1, v_1\}$ will not define an edge
- Many data structures can implement abstract undirected graphs
 - Adjacency matrices
 - Adjacency lists